

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1104

COURSE OUTCOME COVERED

***CO1:** Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS:

Total Marks:40

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I, Judy Allen J (192520055) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Judy Allen J

TECHNICAL PRESENTATION

CSA1104 – Object Oriented Analysis and Design

Presented By,

Judy Allen J

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Modeling Object Relationships in a University Management System

Introduction

- A University Management System (UMS) automates academic and administrative activities.
- It manages Students, Faculty, Departments, Courses, and Assignments.
- Object-Oriented Design (OOD) models these entities as objects with well-defined relationships.

Problem Statement

- Incorrect object relationships increase software complexity.
- Poor design reduces flexibility and code reusability.
- Proper relationship modeling improves software quality and maintenance.

Types of Object Relationships

Association

- Represents a connection between two independent objects.
- Objects communicate without ownership.

Example: Student enrolls in Course.

Aggregation

- Represents a weak "Has-A" relationship.
- Parent and child objects exist independently.

Example: Department has Faculty members.

Types of Object Relationships

Composition

- Represents a strong "Part-Of" relationship.
- Child objects cannot exist without the parent.

Example: Course contains Assignments.

UML Class Diagram

University Management System UML

Classes

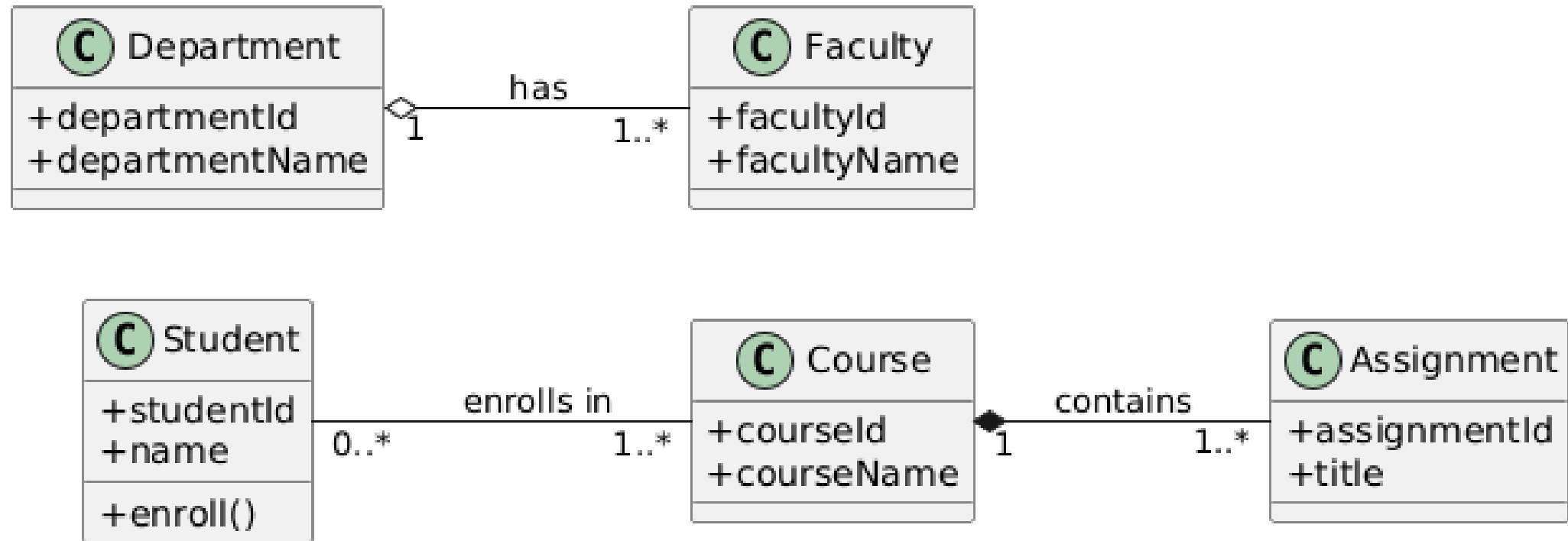
- Student
- Faculty
- Department
- Course
- Assignment

Relationships

- Student enrolls in Course (*Association*)
- Department has Faculty (*Aggregation*)
- Course contains Assignment (*Composition*)

UML Class Diagram

University Management System - Object Relationships



— Association

◇ Aggregation

◆ Composition

Analysis & Benefits

Analysis of Object Relationships

Association

- Supports interaction between independent objects.
- Reduces unnecessary dependencies.

Aggregation

- Promotes object sharing and code reuse.
- Allows independent object lifecycle.

Composition

- Provides strong ownership and data integrity.
- Simplifies object lifecycle management.

Analysis & Benefits

Overall Benefits

- ✓ Improves Flexibility
- ✓ Enhances Reusability
- ✓ Simplifies Maintenance
- ✓ Supports Scalability

Conclusion

- Object relationship modeling is essential in Object-Oriented Design. Association, Aggregation, and Composition define efficient interactions between objects.
- Proper relationship modeling improves software flexibility, maintainability, and scalability.
- Well-designed relationships lead to reliable and high-quality University Management Systems.

References

- Bennett, S., McRobb, S., & Farmer, R. (2023). *Object-oriented systems analysis and design using UML* (5th ed.). McGraw-Hill Education.
- Pressman, R. S., & Maxim, B. R. (2024). *Software engineering: A practitioner's approach* (10th ed.). McGraw-Hill.
- Object Management Group. (n.d.). *Unified Modeling Language (UML®)*. Retrieved July 26, 2026